# Grade 7 Computing

## Strand 1: Introduction to Computing

### Lesson 1 — Components of Computers and File Management

**Codes:** B7.1.1.1.1 – B7.1.1.1.4, B7.1.1.2.1 – B7.1.1.2.2 **Format:** Theory Questions and Answers

## Preamble (pre-lesson)

Before the questions, here is the big picture.

A **computer** is a machine that takes data in, works on it, and gives results out.

```
[INPUT] ■■■ [COMPUTER / STORAGE] ■■■ [OUTPUT]
keyboard the machine itself monitor
mouse keeps your files printer
scanner runs the programs speakers
```

Everything below is built on this one idea: **in** → **process/store** → **out**.

## A. Generations of computers

**Q1.** What is a computer? **Ans.** A computer is a machine that takes data in, processes it, and gives results out.

**Q2.** What is hardware? **Ans.** Hardware is any physical part of a computer you can touch.

**Q3.** What is software? **Ans.** Software is the set of programs that tell the hardware what to do.

**Q4.** What is a generation of computers? **Ans.** A generation is a stage in computer history marked by the technology used to build it.

**Q5.** How many main generations of computers are there? **Ans.** There are four main generations.

**Q6.** What technology marked the first generation of computers? **Ans.** The first generation used vacuum tubes.

**Q7.** What technology marked the second generation of computers? **Ans.** The second generation used transistors.

**Q8.** What technology marked the third generation of computers? **Ans.** The third generation used integrated circuits (chips).

**Q9.** What technology marked the fourth generation of computers? **Ans.** The fourth generation used the microprocessor.

**Q10.** What is a microprocessor? **Ans.** A microprocessor is a whole central processing unit built onto a single small chip.

**Q11.** Which generation do today's laptops and phones belong to? **Ans.** Today's laptops and phones belong to the fourth generation.

**Q12.** Why is the fourth generation important? **Ans.** It is important because it made computers small, cheap, and fast enough for everyday use.

## B. Input devices

**Q13.** What is an input device? **Ans.** An input device is a tool that puts data into a computer.

**Q14.** Mention five examples of input devices. **Ans.**

- Keyboard
- Mouse

- Barcode scanner
- Microphone
- Webcam

**Q15.** What does a keyboard do? **Ans.** A keyboard lets you type letters, numbers, and commands into the computer.

**Q16.** What does a mouse do? **Ans.** A mouse lets you point at and select things on the screen.

**Q17.** What does a barcode scanner do? **Ans.** A barcode scanner reads the striped code printed on a product.

**Q18.** Where do people use barcode scanners? **Ans.** People use barcode scanners in shops, supermarkets, and libraries.

**Q19.** What does a QR code scanner do? **Ans.** A QR code scanner reads a square black-and-white code to open information quickly.

**Q20.** What does a microphone do? **Ans.** A microphone captures sound and sends it into the computer.

**Q21.** What does a webcam do? **Ans.** A webcam captures pictures and video for the computer.

**Q22.** What does a scanner do? **Ans.** A scanner copies a paper document into a digital file.

**Q23.** What is a touchscreen? **Ans.** A touchscreen is a screen that also accepts taps and touches as input.

**Q24.** State the types of common input devices used in a school computer lab. **Ans.**

- Keyboard
- Mouse
- Scanner
- Microphone
- Webcam

## C. Output devices

**Q25.** What is an output device? **Ans.** An output device is a tool that shows or produces the results from a computer.

**Q26.** Mention five examples of output devices. **Ans.**

- Monitor
- Printer
- Speakers
- Graphing plotter
- Data projector

**Q27.** What does a monitor do? **Ans.** A monitor displays pictures and text you can see.

**Q28.** What does a printer do? **Ans.** A printer produces paper copies of digital documents.

**Q29.** What do speakers do? **Ans.** Speakers produce sound from the computer.

**Q30.** What does a graphing plotter do? **Ans.** A graphing plotter draws large and very precise pictures such as maps and engineering drawings.

**Q31.** What does a data projector do? **Ans.** A data projector shows a large image on a wall or screen.

**Q32.** What does a 3D printer do? **Ans.** A 3D printer builds a solid object from a digital design.

## D. Storage devices

**Q33.** What is storage? **Ans.** Storage is the place where a computer keeps data until it is needed.

**Q34.** What is an external storage device? **Ans.** An external storage device is a storage tool you can plug in and carry with you.

**Q35.** Mention five types of storage devices. **Ans.**

- Full-sized external hard drive
- Hard disk drive (HDD)

- Solid-state drive (SSD)
- USB flash drive
- Memory card

**Q36.** What is a hard disk drive (HDD)? **Ans.** A hard disk drive is a storage device that uses spinning disks inside it.

**Q37.** What is a solid-state drive (SSD)? **Ans.** A solid-state drive is a storage device with no moving parts that is faster and quieter than a hard disk.

**Q38.** What is a USB flash drive? **Ans.** A USB flash drive is a small portable storage device that plugs into a USB port.

**Q39.** What is a memory card used for? **Ans.** A memory card stores data in cameras, phones, and some computers.

**Q40.** Which is faster, an HDD or an SSD? **Ans.** An SSD is faster than an HDD.

## E. The operating system and Windows

**Q41.** What is an operating system? **Ans.** An operating system is the main software that runs the computer and manages all other programs.

**Q42.** Give one example of an operating system. **Ans.** Windows is one example of an operating system.

**Q43.** What is the Start button in Windows used for? **Ans.** The Start button opens programs, search, settings, and power options.

**Q44.** How do you search for a program in Windows? **Ans.** You click Start and type the name of the program.

**Q45.** Where do you change the wallpaper in Windows? **Ans.** You change the wallpaper in Settings under Personalisation.

**Q46.** What does the Power option in Windows do? **Ans.** The Power option lets you Sleep, Shut down, or Restart the computer.

**Q47.** What does "Shut down" do? **Ans.** Shut down turns the computer completely off.

**Q48.** What does "Restart" do? **Ans.** Restart turns the computer off and immediately back on again.

## F. Files and folders

**Q49.** What is a file? **Ans.** A file is a named collection of data such as a document, photo, or video.

**Q50.** What is a folder? **Ans.** A folder is a container that holds files and other folders.

**Q51.** What is file management? **Ans.** File management is the skill of creating, naming, moving, and finding your files.

**Q52.** Why should a file have a meaningful name? **Ans.** A meaningful name helps you find the file again later.

**Q53.** Give one good example of a file name. **Ans.** Maths_Homework_Week3.docx is one good example of a file name.

**Q54.** Give one bad example of a file name. **Ans.** doc1.docx is one bad example of a file name.

**Q55.** How do you create a new folder? **Ans.** You right-click, choose New, and then choose Folder.

**Q56.** How do you rename a file? **Ans.** You right-click the file and choose Rename.

**Q57.** What is the keyboard shortcut for Copy? **Ans.** The keyboard shortcut for Copy is Ctrl + C.

**Q58.** What is the keyboard shortcut for Paste? **Ans.** The keyboard shortcut for Paste is Ctrl + V.

**Q59.** What is the keyboard shortcut for Cut? **Ans.** The keyboard shortcut for Cut is Ctrl + X.

**Q60.** What is the difference between Cut and Copy? **Ans.** Cut removes the file from its original place while Copy leaves the original in place.

**Q61.** What happens when you press Delete on a file? **Ans.** The file goes to the Recycle Bin.

**Q62.** How do you restore a file from the Recycle Bin? **Ans.** You open the Recycle Bin, right-click the file, and choose Restore.

**Q63.** How do you search for a lost file in Windows? **Ans.** You open Start and type the file name in the search box.

## Text diagram — how the parts connect

```
+----------------------+
keyboard ----| |
mouse -------| COMPUTER (CPU) |----- monitor
scanner -----| + STORAGE inside |----- printer
webcam ------| |----- speakers
microphone --| |----- projector
+----------------------+
|
+---- external storage
(HDD / SSD / USB)
```

## Exercises (8 sections × 5 questions)

*Answers are given in full after each section. Questions marked **[Review]** are spiral questions from earlier work in this lesson.*

### Exercise 1 — Generations and basic terms

1. What is a computer?
2. What is hardware?
3. What is software?
4. Which generation uses the microprocessor?
5. **[Review]** What is an input device?

**Answers**

1. A computer is a machine that takes data in, processes it, and gives results out.
2. Hardware is any physical part of a computer you can touch.
3. Software is the set of programs that tell the hardware what to do.
4. The fourth generation uses the microprocessor.
5. An input device is a tool that puts data into a computer.

### Exercise 2 — Input devices

1. What is an input device?
2. Mention four examples of input devices.
3. What does a barcode scanner do?
4. What does a microphone do?
5. **[Review]** What is a webcam?

**Answers**

1. An input device is a tool that puts data into a computer.
2. Keyboard, mouse, scanner, microphone.
3. A barcode scanner reads the striped code printed on a product.
4. A microphone captures sound and sends it into the computer.
5. A webcam captures pictures and video for the computer.

### Exercise 3 — Output devices

1. What is an output device?
2. Mention four examples of output devices.
3. What does a graphing plotter do?
4. What do speakers do?
5. **[Review]** What does a monitor do?

**Answers**

1. An output device is a tool that shows or produces the results from a computer.
2. Monitor, printer, speakers, projector.
3. A graphing plotter draws large and very precise pictures such as maps and engineering drawings.
4. Speakers produce sound from the computer.
5. A monitor displays pictures and text you can see.

## Exercise 4 — Storage devices

1. What is storage?
2. Mention four types of storage devices.
3. What is the difference between an HDD and an SSD?
4. What is a USB flash drive?
5. **[Review]** Which is faster, an HDD or an SSD?

**Answers**

1. Storage is the place where a computer keeps data until it is needed.
2. External hard drive, HDD, SSD, USB flash drive.
3. An HDD uses spinning disks while an SSD has no moving parts.
4. A USB flash drive is a small portable storage device that plugs into a USB port.
5. An SSD is faster than an HDD.

## Exercise 5 — Operating system

1. What is an operating system?
2. Give one example of an operating system.
3. What is the Start button used for?
4. How do you search for a program in Windows?
5. **[Review]** What does the Power option do?

**Answers**

1. An operating system is the main software that runs the computer and manages all other programs.
2. Windows is one example.
3. The Start button opens programs, search, settings, and power options.
4. You click Start and type the name of the program.
5. The Power option lets you Sleep, Shut down, or Restart the computer.

## Exercise 6 — Files and folders

1. What is a file?
2. What is a folder?
3. Why should a file have a meaningful name?
4. Give one good example of a file name.
5. **[Review]** How do you create a new folder?

**Answers**

1. A file is a named collection of data such as a document, photo, or video.
2. A folder is a container that holds files and other folders.
3. A meaningful name helps you find the file again later.
4. Maths_Homework_Week3.docx.
5. You right-click, choose New, and then choose Folder.

## Exercise 7 — Move, copy, delete

1. What is the keyboard shortcut for Copy?
2. What is the keyboard shortcut for Paste?
3. What is the difference between Cut and Copy?

4. What happens when you press Delete on a file?
5. **[Review]** How do you rename a file?

**Answers**

1. Ctrl + C.
2. Ctrl + V.
3. Cut removes the file from its original place while Copy leaves the original in place.
4. The file goes to the Recycle Bin.
5. You right-click the file and choose Rename.

### Exercise 8 — Mixed and scenario

1. A student saves work as "doc1.docx". Why is this a problem?
2. Give a better name for a JHS 1 computing project about file management.
3. How do you find a lost file in Windows?
4. **[Review]** What is a USB flash drive?
5. **[Review]** Mention four output devices.

**Answers**

1. The name does not describe the content so the file is hard to find later.
2. JHS1_Computing_FileManagement_Report.docx.
3. You open Start and type the file name in the search box.
4. A USB flash drive is a small portable storage device that plugs into a USB port.
5. Monitor, printer, speakers, projector.

## Summary table

| Term | One-line meaning |
|---|---|
| Input device | puts data into the computer |
| Output device | shows results out of the computer |
| Storage | keeps data until needed |
| 4th generation | microprocessor era |
| Operating system | main software that runs the computer |
| File | named collection of data |
| Folder | container that holds files |

*NaCCA CCP B7.1.1 — Theory Questions and Answers format. Plain simple English. British notation.*